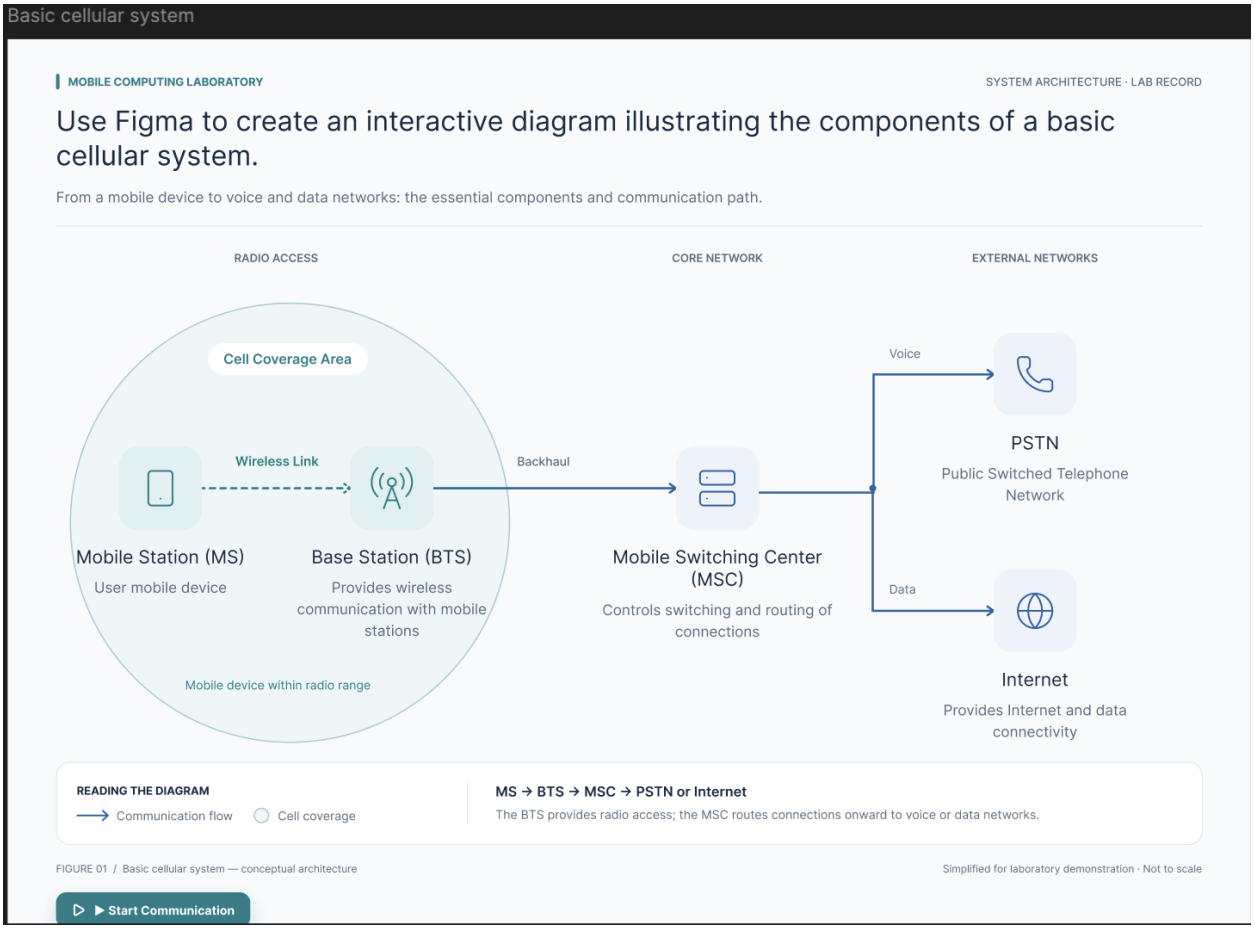
EXPERIMENT-1

MOBILE COMPUTING



MOBILE COMPUTING LABORATORY

SYSTEM ARCHITECTURE · LAB RECORD

CELLULAR SYSTEM COMPONENTS

The five essential components that connect a mobile device to voice and data networks.

RADIO ACCESS

Mobile Station (MS)

User mobile device that communicates with the cellular network.

RADIO ACCESS

Base Station (BTS)

Provides wireless communication between mobile stations and the network.

CORE NETWORK

Mobile Switching Center (MSC)

Controls switching and routing of connections.

EXTERNAL NETWORKS

PSTN

Public Switched Telephone Network for voice communication.

EXTERNAL NETWORKS

Internet

Provides Internet and data connectivity.

The BTS provides radio access; the MSC routes connections onward to voice or data networks.

← Back to Diagram

MS → BTS → MSC → PSTN or Internet

Radio access → Core network → External networks

FIGURE 02 / Cellular system components — functions and roles

Simplified for laboratory demonstration · Not to scale

Cellular communication flow

MOBILE COMPUTING LABORATORY

SYSTEM ARCHITECTURE · LAB RECORD

CELLULAR COMMUNICATION FLOW

Follow the communication path from radio access through the core network to voice or data services.



← Back to Diagram

MS → BTS → MSC → PSTN or Internet

Radio access → Core network → External networks

FIGURE 03 / Cellular communication flow — voice and data routing

Simplified for laboratory demonstration · Not to scale